4th

May 7, 2026

Create a Linear Regression Model using Python/R to predict home prices using Boston Housing Dataset (https://www.kaggle.com/c/boston-housing). The Boston Housing dataset contains information about various houses in Boston through different parameters. There are 506 samples and 14 feature variables in this dataset.

[ ]:
```
import numpy as np
import pandas as pd
from sklearn.datasets import fetch_california_housing
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_squared_error, r2_score
from sklearn.preprocessing import StandardScaler
import seaborn as sns
import matplotlib.pyplot as plt
```

[2]:
```
# Load dataset
housing_data = fetch_california_housing()

housing_df = pd.DataFrame(housing_data.data, columns=housing_data.feature_names)
```

[3]:
```
housing_df.head()
```

[3]:
```
   MedInc  HouseAge  AveRooms  AveBedrms  Population  AveOccup  Latitude  \
0  8.3252      41.0  6.984127   1.023810       322.0  2.555556     37.88
1  8.3014      21.0  6.238137   0.971880      2401.0  2.109842     37.86
2  7.2574      52.0  8.288136   1.073446       496.0  2.802260     37.85
3  5.6431      52.0  5.817352   1.073059       558.0  2.547945     37.85
4  3.8462      52.0  6.281853   1.081081       565.0  2.181467     37.85

   Longitude
0    -122.23
1    -122.22
2    -122.24
3    -122.25
4    -122.25
```

[4]:
```
print("Shape and Size: ", housing_df.shape, housing_df.size)
housing_df['PRICE'] = housing_data.target
```

```
Shape and Size:  (20640, 8) 165120
```

[5]:
```python
# Basic Info
print("Shape:", housing_df.shape)
print("\nColumns:\n", housing_df.columns)
```

```
Shape: (20640, 9)

Columns:
 Index(['MedInc', 'HouseAge', 'AveRooms', 'AveBedrms', 'Population', 'AveOccup',
       'Latitude', 'Longitude', 'PRICE'],
      dtype='object')
```

[6]:
```python
print("\nSummary:\n")
housing_df.describe()
```

```
Summary:
```

[6]:

	MedInc	HouseAge	AveRooms	AveBedrms	Population	\
count	20640.000000	20640.000000	20640.000000	20640.000000	20640.000000	
mean	3.870671	28.639486	5.429000	1.096675	1425.476744	
std	1.899822	12.585558	2.474173	0.473911	1132.462122	
min	0.499900	1.000000	0.846154	0.333333	3.000000	
25%	2.563400	18.000000	4.440716	1.006079	787.000000	
50%	3.534800	29.000000	5.229129	1.048780	1166.000000	
75%	4.743250	37.000000	6.052381	1.099526	1725.000000	
max	15.000100	52.000000	141.909091	34.066667	35682.000000	

	AveOccup	Latitude	Longitude	PRICE
count	20640.000000	20640.000000	20640.000000	20640.000000
mean	3.070655	35.631861	-119.569704	2.068558
std	10.386050	2.135952	2.003532	1.153956
min	0.692308	32.540000	-124.350000	0.149990
25%	2.429741	33.930000	-121.800000	1.196000
50%	2.818116	34.260000	-118.490000	1.797000
75%	3.282261	37.710000	-118.010000	2.647250
max	1243.333333	41.950000	-114.310000	5.000010

[7]:
```python
# Introduce missing values
housing_df.loc[housing_df.sample(frac=0.05).index, 'MedInc'] = np.nan
# Check missing values
print("\nMissing Values:\n", housing_df.isnull().sum())
```

```
Missing Values:
 MedInc        1032
```

```
HouseAge         0
AveRooms         0
AveBedrms        0
Population       0
AveOccup         0
Latitude         0
Longitude        0
PRICE            0
dtype: int64
```

[8]:
```python
# Handling missing values (mean imputation)
housing_df.fillna(housing_df.mean(), inplace=True)
# Check missing values
print("\nMissing Values:\n", housing_df.isnull().sum())
```

```
Missing Values:
 MedInc        0
HouseAge      0
AveRooms      0
AveBedrms     0
Population    0
AveOccup      0
Latitude      0
Longitude     0
PRICE         0
dtype: int64
```

[9]:
```python
housing_df.head()
```

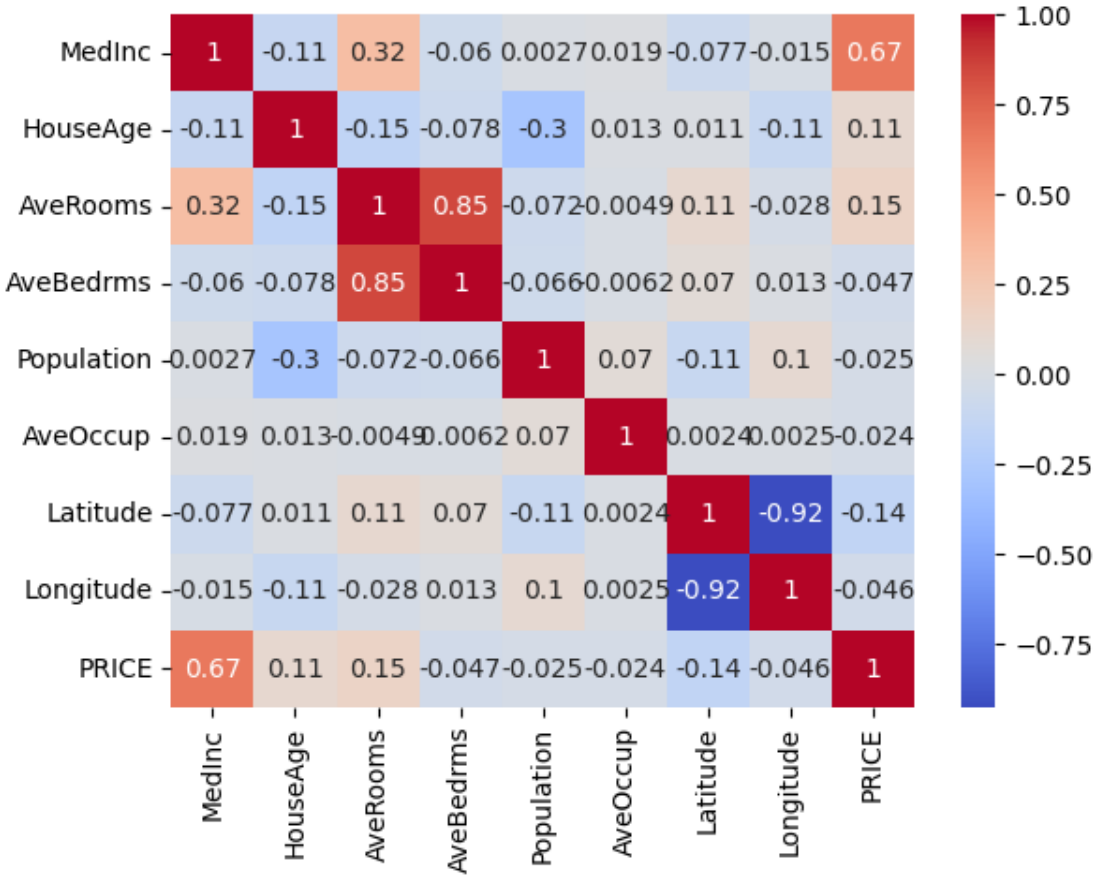
[9]:

	MedInc	HouseAge	AveRooms	AveBedrms	Population	AveOccup	Latitude	\
0	8.3252	41.0	6.984127	1.023810	322.0	2.555556	37.88	
1	8.3014	21.0	6.238137	0.971880	2401.0	2.109842	37.86	
2	7.2574	52.0	8.288136	1.073446	496.0	2.802260	37.85	
3	5.6431	52.0	5.817352	1.073059	558.0	2.547945	37.85	
4	3.8462	52.0	6.281853	1.081081	565.0	2.181467	37.85	

	Longitude	PRICE
0	-122.23	4.526
1	-122.22	3.585
2	-122.24	3.521
3	-122.25	3.413
4	-122.25	3.422

[10]:
```python
sns.heatmap(housing_df.corr(), annot=True, cmap='coolwarm')
```

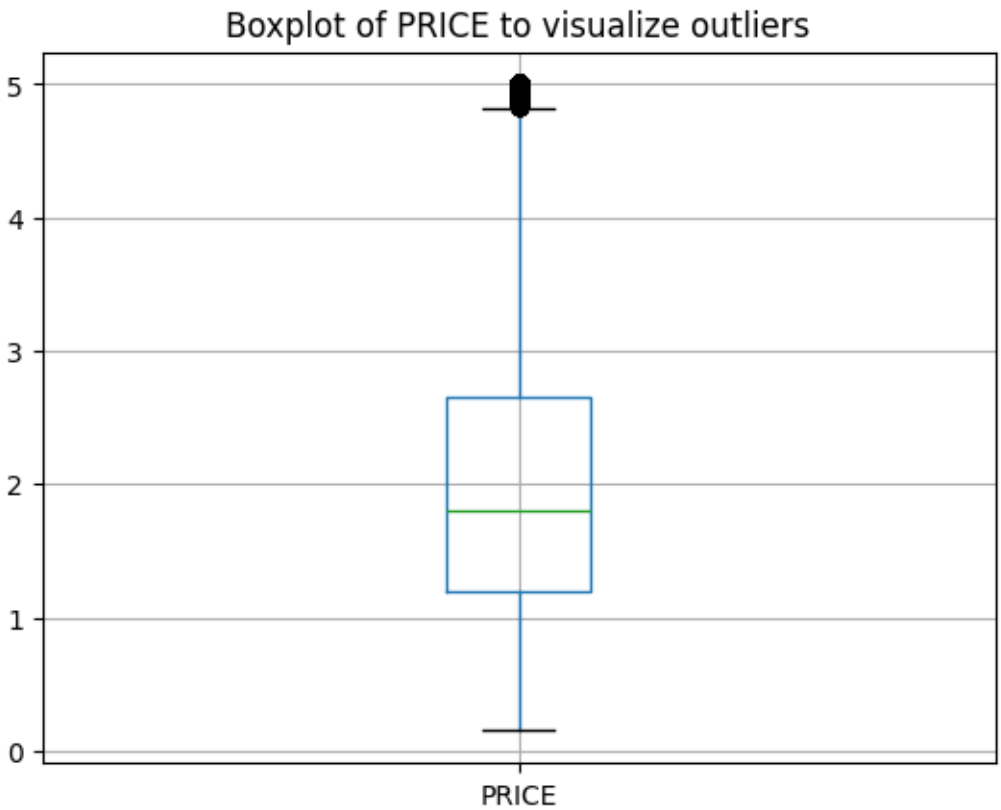
[10]:
```
<Axes: >
```



[ ]:
```python
# Outlier Detection and Removal using IQR method
def detect_outliers_iqr(data, column):
    Q1 = data[column].quantile(0.25)
    Q3 = data[column].quantile(0.75)
    IQR = Q3 - Q1
    lower_bound = Q1 - 1.5 * IQR
    upper_bound = Q3 + 1.5 * IQR
    outliers = data[(data[column] < lower_bound) | (data[column] > upper_bound)]
    return outliers, lower_bound, upper_bound

# Detect outliers in the PRICE column
outliers_price, lower_bound, upper_bound = detect_outliers_iqr(housing_df,␣
 ↪'PRICE')
print("Number of outliers in PRICE:", len(outliers_price))
print("Outliers in PRICE:")
print(outliers_price.head())  # Show first few outliers
print(f"Lower bound: {lower_bound}, Upper bound: {upper_bound}")

# Visualize outliers using boxplot
```

```python
housing_df.boxplot(column='PRICE')
plt.title('Boxplot of PRICE to visualize outliers')
plt.show()

# Remove outliers
housing_df_clean = housing_df[(housing_df['PRICE'] >= lower_bound) &
 ↪(housing_df['PRICE'] <= upper_bound)]
print("Original dataset shape:", housing_df.shape)
print("Cleaned dataset shape:", housing_df_clean.shape)
print("Removed", len(housing_df) - len(housing_df_clean), "outliers")

# Update variables for cleaned data
X_clean = housing_df_clean.drop("PRICE", axis=1)
y_clean = housing_df_clean["PRICE"]
```

```
Number of outliers in PRICE: 1071
Outliers in PRICE:
       MedInc  HouseAge  AveRooms  AveBedrms  Population  AveOccup  Latitude  \
89   1.243400      52.0  2.929412   0.917647       396.0  4.658824     37.80
140  6.362400      30.0  5.615385   0.730769       126.0  2.423077     37.81
459  3.869035      52.0  2.436000   0.944000      1349.0  5.396000     37.87
489  3.041700      48.0  4.690632   1.126362      1656.0  3.607843     37.86
493  7.852100      52.0  7.794393   1.051402       517.0  2.415888     37.86

     Longitude    PRICE
89     -122.27  5.00001
140    -122.18  4.83300
459    -122.25  5.00001
489    -122.25  4.89600
493    -122.24  5.00001
Lower bound: -0.9808749999999995, Upper bound: 4.824124999999999
```




```
Original dataset shape: (20640, 9)
Cleaned dataset shape: (19569, 9)
Removed 1071 outliers
```

[13]:
```
# Feature & Target split
X = housing_df.drop("PRICE", axis=1)
y = housing_df["PRICE"]
```

[14]:
```
#Splitting data into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=10
)
```

[15]:
```
# Feature Scaling
scaler = StandardScaler()

X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
```

[16]:
```
# Model Training
model = LinearRegression()
```

```
model.fit(X_train_scaled, y_train)
```

[16]: LinearRegression()

[17]:
```
# Predictions
y_pred = model.predict(X_test_scaled)
```

[18]:
```
# Evaluation
rmse = np.sqrt(mean_squared_error(y_test, y_pred))
r2 = r2_score(y_test, y_pred)

print("\n==============================")
print("Model Performance")
print("==============================")
print("RMSE:", rmse)
print("R2 Score:", round(r2, 2))
```

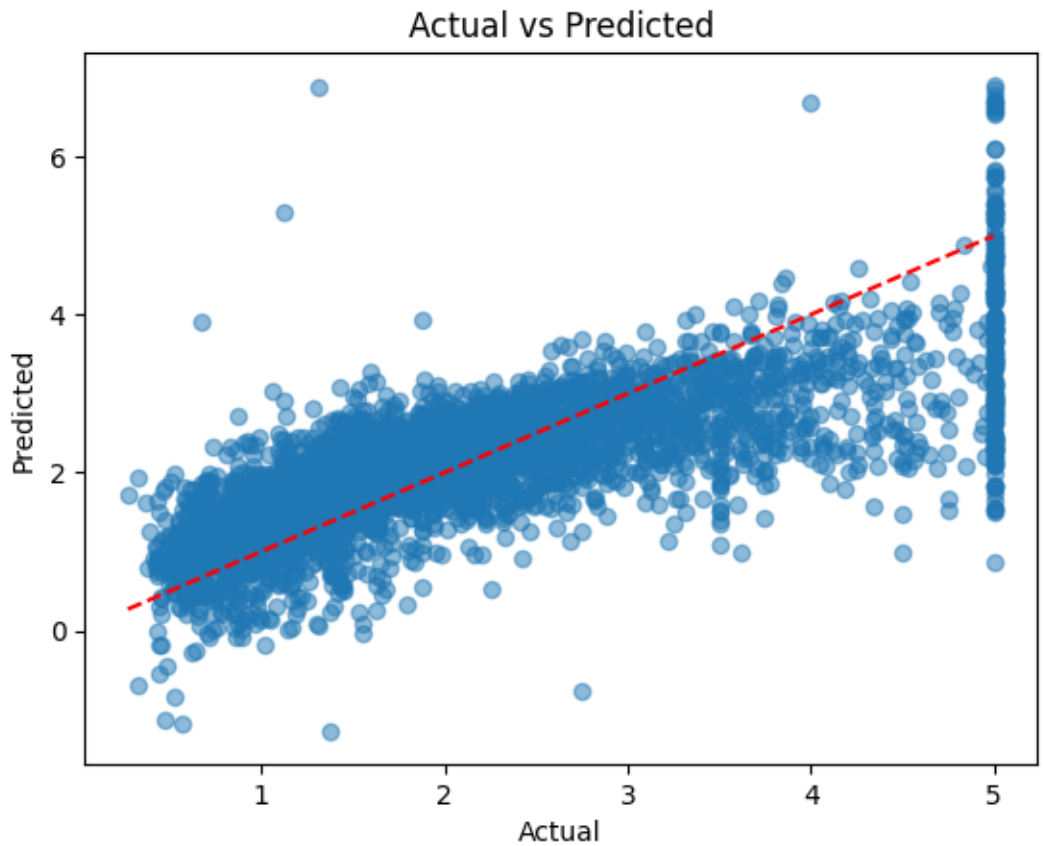
```
==============================
Model Performance
==============================
RMSE: 0.7662264057427334
R2 Score: 0.57
```

[19]:
```
# Cleaned dataset model training + evaluation
X_train_clean, X_test_clean, y_train_clean, y_test_clean = train_test_split(
    X_clean, y_clean, test_size=0.2, random_state=10
)

scaler_clean = StandardScaler()
X_train_clean_scaled = scaler_clean.fit_transform(X_train_clean)
X_test_clean_scaled = scaler_clean.transform(X_test_clean)

model_clean = LinearRegression()
model_clean.fit(X_train_clean_scaled, y_train_clean)

y_pred_clean = model_clean.predict(X_test_clean_scaled)

rmse_clean = np.sqrt(mean_squared_error(y_test_clean, y_pred_clean))
r2_clean = r2_score(y_test_clean, y_pred_clean)

print("\n==============================")
print("Model Performance on Cleaned Data")
print("==============================")
print("RMSE:", rmse_clean)
print("R2 Score:", round(r2_clean, 2))
```

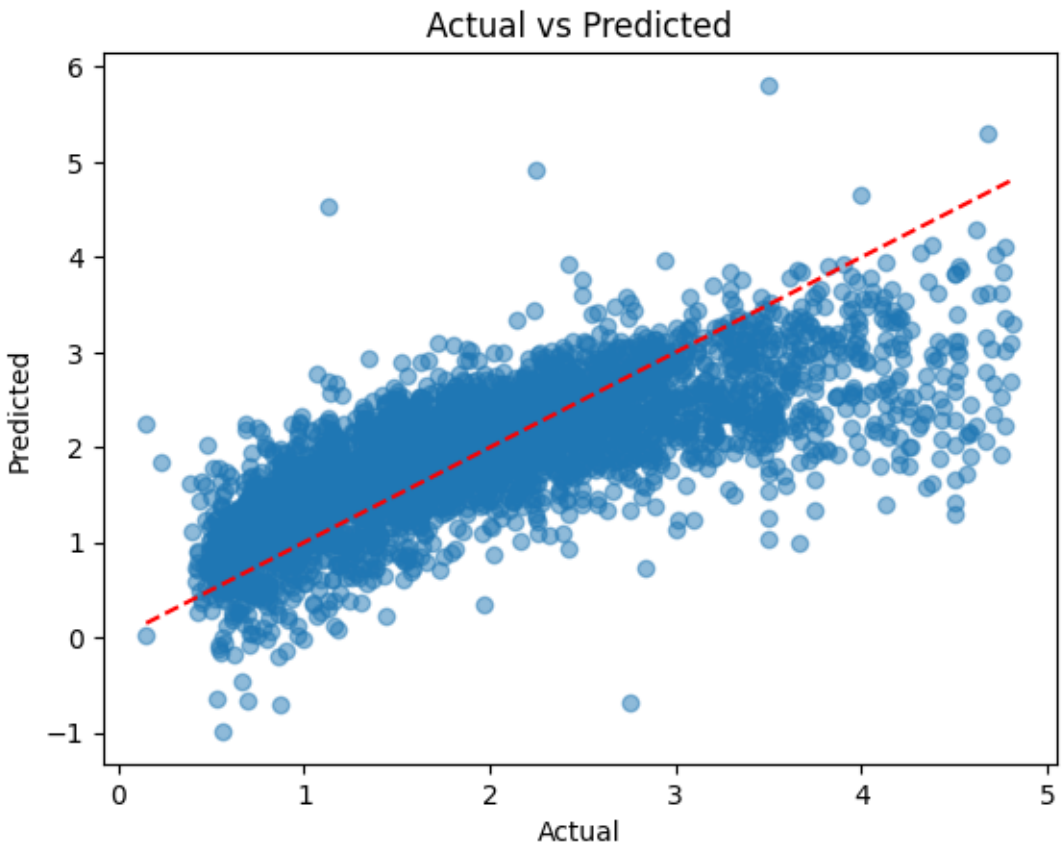
```
print("\n==============================")
print("Comparison")
print("==============================")
print("Original RMSE:", rmse, "Cleaned RMSE:", rmse_clean)
print("Original R2:", round(r2, 2), "Cleaned R2:", round(r2_clean, 2))
```

```
==============================
Model Performance on Cleaned Data
==============================
RMSE: 0.6368822083671271
R2 Score: 0.55

==============================
Comparison
==============================
Original RMSE: 0.7662264057427334 Cleaned RMSE: 0.6368822083671271
Original R2: 0.57 Cleaned R2: 0.55
```

[22]:
```
plt.scatter(y_test, y_pred, alpha=0.5)
plt.xlabel("Actual")
plt.ylabel("Predicted")
plt.title("Actual vs Predicted")
plt.plot([min(y_test), max(y_test)], [min(y_test), max(y_test)], 'r--')
plt.show()
```



[23]:
```
plt.scatter(y_test_clean, y_pred_clean, alpha=0.5)
plt.xlabel("Actual")
plt.ylabel("Predicted")
plt.title("Actual vs Predicted")
plt.plot([min(y_test_clean), max(y_test_clean)], [min(y_test_clean),␣
 ↪max(y_test_clean)], 'r--')
plt.show()
```


Actual vs Predicted
Predicted
Actual
6
5
4
3
2
1
0
−1
0
1
2
3
4
5